

Ivan Antony Babu

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OBJECTIVE

Robotics and Embedded Systems Engineer with strong interests in UAVs, Autonomous Systems, Software-Defined Radio, Wireless Communication, Computer Vision, and AI-driven technologies. Experienced in developing intelligent hardware-software solutions for real-world engineering applications.

RELEVANT COURSEWORK

- Calculus
- Linear Algebra & Transform Techniques
- Signals & Systems
- Digital Signal Processing
- Digital Image Processing
- Object Oriented Programming in C++
- Data Communications & Networking
- Microprocessors and Microcontrollers
- Embedded Systems
- Cyber Physical Systems
- Control Systems
- Artificial Intelligence & Robotics
- RF Circuit Design

EDUCATION

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| Cochin University of Science and Technology
Bachelor of Technology , Electronics and Communications Engineering
– CGPA-:8.58 | Kochi,India
2023–2027 |
| Indian Institute of Technology Madras
Bachelor of Science (B.Sc), Data Science and Applications
– CGPA-:8.33 | Madras, India
2023–2027 |
| Lord's Academy CMI ICSE School
High School Diploma
– ISC 94% ,ICSE 93% | Thrissur,India
2008–2022 |

SKILLS

Languages: Python, C++, C, Java

OS: Linux, Windows

Tools & Platforms: ROS, Gazebo, Git, GitHub, Arduino, GNU Radio, KiCad, SolidWorks, Matlab, ANSYS HFSS, LTSpice, Proteus, Bash, VS Code, LaTeX, MS Word

Soft Skills: Leadership, Team Building, Project Management, Public Speaking, Analytical Thinking, Fast Learner, Technical Report Writing

Linguistic Proficiency: English(Professional), Malayalam(Bilingual)

PROFESSIONAL EXPERIENCE

[Linked in](#)

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| IIT Tirupati Navavishkar I-Hub Foundation (IITTNiF)
Project Intern
– Developed a custom Indoor Mapping System using ROS & SLAM. | Tirupati, India
Jan 2026 – Apr 2026 |
| Department of Electronics, CUSAT
Summer Intern
– Completed intensive hands-on training on Software-Defined Radio (SDR) platforms, involving GNU Radio, spectrum sensing, and wireless transmission using RTL-SDR and USRP systems. | Kochi, India
Jun 2026 |
| Kerala State Electronics Development Corporation (KELTRON)
Summer Intern
– Designed and implemented a Library Management System using Python, SQL, and Tkinter. | Kochi, India
May 2024 |

PROJECTS

[GitHub](#)

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| Indoor Mapping and Navigation System using Raspberry Pi 5
<i>Research Project — IIT Tirupati Navavishkar i-Hub Foundation (IITTNiF) & CUSAT</i> <ul style="list-style-type: none">• Developed a low-cost indoor localization and mapping system using Raspberry Pi 5, monocular camera, IMU, and Time-of-Flight (ToF) LiDAR sensors.• Implemented Visual-Inertial SLAM with multi-sensor fusion to generate real-time 2D occupancy maps for GPS-denied indoor environments. | 2026 |
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- Designed a QR-code-based navigation framework enabling users to access interactive indoor maps and shortest-path navigation through a web interface.
- Improved localization accuracy by integrating ToF-based range validation, significantly reducing trajectory drift during indoor mapping experiments.

Audio-Band FSK Text Modem using GNU Radio

2026

SDR Project - DOE, CUSAT

- Designed and implemented an audio-band Frequency Shift Keying (FSK) modem using GNU Radio to transmit text data through standard computer microphones and speakers.
- Developed complete transmitter and receiver flowgraphs incorporating modulation, demodulation, quadrature demodulation, binary slicing, and file reconstruction.
- Implemented reliable end-to-end text communication by converting digital data into audio tones and recovering the transmitted message in real time.
- Evaluated modem performance under varying channel conditions and optimized parameters to improve decoding accuracy and communication reliability.

SDR-Based Anti-Jamming Audio Communication System

2026

SDR Project - DOE, CUSAT

- Developed a full-duplex Software Defined Radio communication system using the Ettus B210 to investigate interference mitigation in wireless audio transmission.
- Simulated intentional wideband noise jamming while transmitting Narrowband FM (NBFM) audio over independent SDR channels.
- Designed adaptive software filtering techniques to suppress interference and recover intelligible audio under high jammer power conditions.
- Analyzed receiver performance by varying jammer intensity, filter bandwidth, and signal-to-noise ratio (SNR) to evaluate anti-jamming effectiveness.

Smart Attendance and Student Monitoring System (SAMS)

2025

Embedded Systems & IoT Project - CUSAT

- Designed an intelligent classroom monitoring system using Raspberry Pi 5, ESP8266, RFID authentication, and computer vision.
- Developed automated attendance logging through RFID identification integrated with a centralized Flask-based server.
- Implemented real-time student engagement monitoring using facial landmark detection and Eye Aspect Ratio (EAR) based attention analysis.

Smart Fence Intrusion Detection System

2024

Embedded Systems & IoT Project - CUSAT

- Developed an intelligent perimeter security system capable of detecting intrusion events using multiple environmental sensors and embedded controllers.

PUBLICATIONS (SELECTED)

 Google Scholar

- **Ivan Antony Babu**, “A Low-Cost ToF LiDAR-Assisted Visual-Inertial SLAM System Using Multi-Sensor Fusion for Indoor Localization on Raspberry Pi 5,” *International Conference on Trends in Engineering Systems and Technologies (ICTEST)*, Kochi, India, 2026 (Accepted).

LEADERSHIP EXPERIENCE

 LinkedIn

AROHA Drone Club

Kochi, India

Team Lead (Previously Team Manager, Electronics Member)

Mar 2025 – Present

- Led AROHA to national and international competition success, achieving World Rank 15 and Kerala Rank 1 at ISDC, alongside AIR 9 and Kerala Rank 1 at SAE Aerothon.

IEEE CUSAT Student Branch Chapter

Kochi, India

Founding Chair, Aerospace and Electronic Systems Society (AESS)

Mar 2026 – Present

- Spearheaded the establishment and official launch of the AESS chapter at CUSAT. Currently driving technical initiatives, strategic planning, and hands-on workshops tailored to autonomous flight systems and avionics architecture.

– Organized numerous professional workshops and technical events, including PCB Assembly tutorials and drone development bootcamps.

INVITED TALKS

1. “WINGS WITHOUT PILOTS: Entering the UAV Era,” hosted by IEEE IA/IE/PELS Joint Chapter Kerala Section in collaboration with IEEE SB AJCE, IEEE MACE Student Branch, and IEEE CUSAT SB, 2026.

CERTIFICATIONS

Computer Architecture and Organization

NPTEL · 2025

AWS Certifications in Python

AWS — DevTown · 2023

Matlab ONRAMP Certification

MATLAB · 2025

VLSI SoC Design using Verilog HDL certification

Maven Silicon · 2023

AWARDS AND ACHIEVEMENTS

- IIT Tirupati iHub **STAR-PNT Fellowship** Awardee
- Secured **Top 1% rank** in the NPTEL Computer Architecture & Organization course offered by IIT Kharagpur.
- Secured **3rd place** at IDEATHON for AI powered drone farming solution
- ISDC 2026 – **World Rank 15**, Kerala Rank 1 in the International Space Drone Challenge.
- AeroTHON 2025 – **AIR 9**, Kerala Rank 1 in SAE India’s national UAV competition.
- National Science Olympiad **School Topper** (2020, 2022)

REFERENCES

Prof. (Dr.) Binu Paul

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